

TD01 BOOSTRAP LINUX:

```
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ pwd
/workspaces/esilv-marketpulse-g01-t07
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ ls
CONTRIBUTING.md  TEAM.md  data  readiness
README.md        TEAM_TEMPLATE.md  evidence  requirements.txt
'TD 1_2.pdf'     config  labs  src
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ ls -la
total 944
drwxrwxrwx+ 9 codespace root 4096 Oct 7 19:34 .
drwxr-xrwx+ 5 codespace root 4096 Oct 7 19:34 ..
drwxrwxrwx+ 8 codespace root 4096 Oct 7 19:35 .git
-rw-rw-rw- 1 codespace root 178 Oct 7 19:34 .gitignore
-rw-rw-rw- 1 codespace root 2943 Oct 7 19:34 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 10270 Oct 7 19:34 README.md
-rw-rw-rw- 1 codespace root 894815 Oct 7 19:34 'TD 1_2.pdf'
-rw-rw-rw- 1 codespace root 1427 Oct 7 19:34 TEAM.md
-rw-rw-rw- 1 codespace root 1446 Oct 7 19:34 TEAM_TEMPLATE.md
drwxrwxrwx+ 2 codespace root 4096 Oct 7 19:34 config
drwxrwxrwx+ 3 codespace root 4096 Oct 7 19:34 data
drwxrwxrwx+ 2 codespace root 4096 Oct 7 19:34 evidence
drwxrwxrwx+ 2 codespace root 4096 Oct 7 19:34 labs
drwxrwxrwx+ 2 codespace root 4096 Oct 7 19:34 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 19:34 requirements.txt
drwxrwxrwx+ 2 codespace root 4096 Oct 7 19:34 src
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ cd data

@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ ls -la
drwxrwxrwx+ 2 codespace root 4096 Oct 7 19:34 src
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ cd data
• @adja07 → /workspaces/esilv-marketpulse-g01-t07/data (adjasy) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data
• @adja07 → /workspaces/esilv-marketpulse-g01-t07/data (adjasy) $ ls
sample
• @adja07 → /workspaces/esilv-marketpulse-g01-t07/data (adjasy) $ cd sample
• @adja07 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjasy) $ pwd
/workspaces/esilv-marketpulse-g01-t07/data/sample
• @adja07 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjasy) $ ls
bloomberg_reference_expected.json  instruments.json
bloomberg_reference_sample.json    prices.csv
• @adja07 → /workspaces/esilv-marketpulse-g01-t07/data/sample (adjasy) $ cd ../../
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ pwd
/workspaces/esilv-marketpulse-g01-t07
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
  }
}

@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ cat data/sample/instruments.json
{
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
• @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ grep AAPL data/sample/prices.csv
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

1. The instrument is:

Ticker: AAPL
Name: Apple Inc.
Currency: USD
Market: NASDAQ

2. The benchmark is:

Ticker: SP500
Name: S&P 500
Currency: USD
Market: US

3. Their names and tickers are stored in the instruments.json file:
data/sample/instruments.json

```
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep AAPL data/sample/prices.csv
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep SP500 data/sample/prices.csv
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
```

```
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ grep SP500 data/sample/prices.csv
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ python --version
Python 3.14.2
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ git --version
git version 2.55.0
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
```

```

@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ grep SP500 data/sample/prices.csv
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ python --version
Python 3.14.2
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ git --version
git version 2.55.0
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ █

```

The python command src/main.py execute the program in the file main.py

TD02_PYTHON_CSV_JSON:

```

@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ code src/main.py
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },

```

code src/main.py displays the content of the file main.py as indicated below.

```
[Aperçu] README.md  main.py X
src > main.py
1  from pathlib import Path
2  import csv
3  import json
4
5
6  DATA_DIR = Path("data/sample")
7
8  # These starter values mirror config/settings.yml.
9  # settings.yml is a human-readable configuration contract in the CORE.
10 # Parsing YAML is optional and is not required by the 18-hour lab sequence.
11 LOOKBACK_LABEL = "1 month"
12 INTERVAL_LABEL = "Daily"
13
14
```

1. The function `load_instruments()` reads JSON
2. The function `load_prices()` reads csv
3. The function `filter_prices(prices, ticker)` selects one sticker
4. The function `main()` coordinates execution
5. They are stored in `data/sample`

```
58     print(type(instruments))
59     print(instruments.keys())
60     print(instrument["ticker"])
61     print(instrument["name"])
62     print(instrument["currency"])
63
64     print(benchmark["ticker"])
65     print(benchmark["name"])
66
67
```

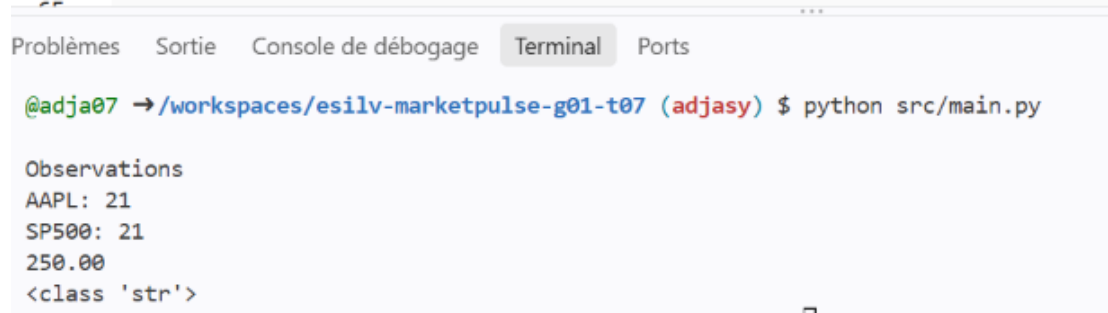
```
Problèmes  Sortie  Console de débogage  Terminal  Ports
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ python src/main.py
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
AAPL
Apple Inc.
USD
SP500
S&P 500
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $
```

```
58     prices = load_prices()
59     print(type(prices))
60     print(type(prices[0]))
61     print(prices[0])
62
63
64 if __name__ == "__main__":
65     main()
66
```

```
Problèmes  Sortie  Console de débogage  Terminal  Ports
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ python src/main.py
AAPL: 21
SP500: 21
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $
```

We can see that prices is a list and that prices[0] is a dictionary.

```
60 |         print(type(prices[0]["close"]))
61 |
62 |
63 | if __name__ == "__main__":
64 |     main()
65 |
```



prices[0][“close”] is a string and we convert it to a float with the command

close = float(prices[0][“close”])

get_first_close(prices) and get_last_close(prices) functions:

```
28 | def get_first_close(prices):
29 |     close = float(prices[0]["close"])
30 |     return close
31 |
32 | def get_last_close(prices):
33 |     close =float(prices[-1]["close"])
34 |     return close
```

```
65     print(prices[0])
66     print(get_first_close(instrument_prices))
67     print(get_first_close(benchmark_prices))
68     print(get_last_close(instrument_prices))
69     print(get_last_close(benchmark_prices))
70
71 if __name__ == "__main__":
72     main()
73
```

Problèmes Sortie Console de débogage Terminal Ports

@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) \$ python src/main.py

```
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
250.0
6600.0
266.2
6742.0
```

@adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjas) \$

display_market_summary function:

```
35 def display_market_summary(asset, prices, show_currency=True):
36     print(asset["ticker"],":", asset["name"])
37     print("Observations:", len(prices[0]))
38     print("First close:", get_first_close(prices), asset["currency"])
39     print("Last close:", get_last_close(prices), asset["currency"])
40
41 def main():
```

```
41 def main():
42     instruments = load_instruments()
43     prices = load_prices()
44
45     instrument = instruments["instrument"]
46     benchmark = instruments["benchmark"]
47
48     instrument_prices = filter_prices(prices, instrument["ticker"],)
49     benchmark_prices = filter_prices(prices, benchmark["ticker"],)
50
51     instrument_latest = instrument_prices[-1]
52     benchmark_latest = benchmark_prices[-1]
53
54     print("=== MarketPulse ===")
55     print()
56     print(f"Period: {LOOKBACK_LABEL}")
57     print(f"Interval: {INTERVAL_LABEL}")
58     print()
59     print("Instrument")
60     display_market_summary(instrument, instrument_prices)
61     print()
62     print("Benchmark")
63     print()
64     display_market_summary(benchmark, benchmark_prices)
65 if __name__ == "__main__":
```

```
● @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ python src/main.py
=== MarketPulse ===

Period: 1 month
Interval: Daily

Instrument
AAPL : Apple Inc.
Observations: 7
First close: 250.0 USD
Last close: 266.2 USD

Benchmark

SP500 : S&P 500
Observations: 7
First close: 6600.0 USD
Last close: 6742.0 USD
○ @adja07 → /workspaces/esilv-marketpulse-g01-t07 (adjasy) $ █
```